

Forged Steel Check Valves

Construction is as follows

Full port or conventional port;

Lift type check valves;

Ball type check valves;

Swing type check valves;

According to requirement equip inside spring;

Bolted bonnet & spiral wound gasket seal bonnet;

Bolted bonnet with spiral-wound gasket, threaded and seal welded bonnet or threaded and pressure seal bonnet;

Socket weld ends to ASME B16.11

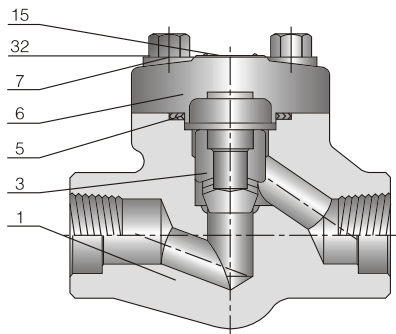
Screwed ends (NPT) to ANSI/ASME B1.20.1.

Disc can change for soft seal disc and ball disc.

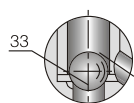


Wanli valves are available in three bonnet designs. The first design is the Bolted Bonnet, with male-female joint, spiral wound gasket, mated in F304L/graphite. Ring joint gasket are also available on request. The second design is the welded bonnet, with a threaded and seal welded joint. On request a full penetration strength welded joint is available, the third design is the pressure seal bonnet, with a threaded and pressure seal bonnet joint.

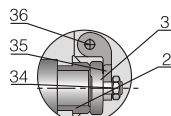
Female Threaded and Socket Welded Check Valves



Please mark in you need load spring



Ball-type check valve schematic



Swing check valve schematic

Application standards

- 1、Design and manufacture conform to: BS5352
- 2、Connection ends conform to:
 - 1) Screw ends dimension conform to ANSI B16.11; JB/T1751
 - 2) Screw ends dimension conform to ANSI B1.20.1; JB/T7306
 - 3) Test and inspection conform to ANSI B16.25; JB/T12224
 - 4) Structure features: ANSI B16.5; JB79
- 3、Test and inspection conform to: API598; GB/T13927; JB/T9092
- 4、Structure features: Bolted bonnet
- 5、Materials conform to ANSI/ASTM.
- 6、Main materials: A105; LF2; F5; F11; F22; 304(L); 316(L); F347; F321; F51; Monel; 20 Alloy.

Carbon steel temperature–pressure rate

CL150–285P.S.I @100° F

CL300–740P.S.I @100° F

CL600–1480P.S.I @100° F

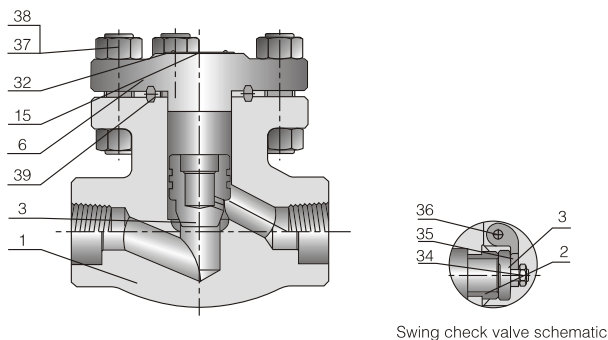
CL800–1975P.S.I @100° F

CL1500–3705P.S.I @100° F

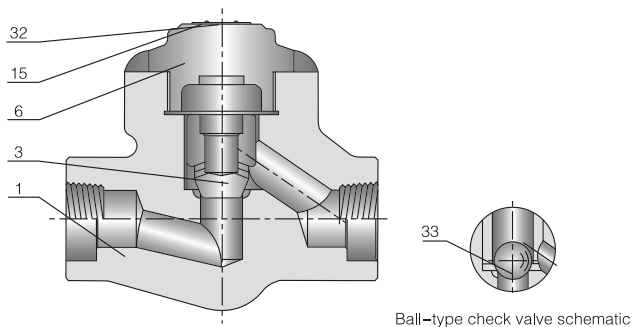
Main part materials

NO.	Part name	A105/F6a	A105/F6aHFS	LF2/304	F11/F6aHF	F304(L)/304(L)	F316(L)/316(L)	F51/F51
1	Body	A105	A105	LF2	F11	F304(L)	F316(L)	F51
2	Seat ring	410	410HF	304	410HF	304(L)	316(L)	F51
3	Disc	F6a	F6a	F304	F6aHF	F304(L)	F316(L)	F51
5	Gasket	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	316+ Flexible graphite	316+ Flexible graphite
6	Bonnet	A105	A105	LF2	F11	F304(L)	F316(L)	F51
7	Bolt	B7	B7	L7	B16	B8(M)	B8(M)	B8M
15	Nameplate	AL	AL	AL	AL	AL	AL	AL
32	Revit	AL	AL	AL	AL	AL	AL	AL
33	Steel ball	430	430	304	STL	F316(L)	316(L)	STL
34	Disc nut	2H	2H	8	8	8(M)	B8(M)	8M
35	Hinge	410	410	304	410	316(L)	316(L)	F51
36	Pin	410	410	304	410	304(L)	316(L)	F51

Female Threaded and Socket Welded Check Valves



To load the springs need to specify in the order form



To load the springs need to specify in the order form

Application standards

- 1、Design and manufacture conform to: BS5352 MSS SP-118
- 2、Connection ends conform to:
 - 1) Screw ends dimension conform to ANSI B16.11; JB/T1751
 - 2) Screw ends dimension conform to ANSI B1.20.1; JB/T7306
 - 3) Test and inspection conform to ANSI B16.25; JB/T12224
 - 4) Structure features: ANSI B16.5; JB79
- 3、Test and inspection conform to: API598; GB/T13927; JB/T9092
- 4、Structure features:
 - Gasket for bonnet connect adopt metal ring(RJ)
 - Bolted bonnet, welded bonnet
- 5、Materials conform to ANSI/ASTM
- 6、Main materials: A105; LF2; F5; F11; F22; 304(L); 316(L); F347; F321; F51; Monel; 20 Alloy.

Carbon steel temperature–pressure rate

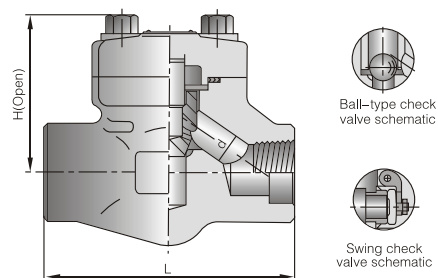
CL150–285 P.S.I @100° F
CL300–740 P.S.I @100° F
CL600–1480 P.S.I @100° F
CL800–1975 P.S.I @100° F
CL1500–3705 P.S.I @100° F
CL2500–6170 P.S.I @100° F

Main part materials

NO.	Part name	A105/F6a	A105/F6aHFS	LF2/304	F11/F6aHF	F304(L)/304(L)	F316(L)/316(L)	F51/F51
1	Body	A105	A105	LF2	F11	F304(L)	F316(L)	F51
2	Sear ring	410	410HF	304	410HF	304(L)	316(L)	F51
3	Disc	F6a	F6a	F304	F6aHF	F304(L)	F316(L)	F51
6	Bonnet	A105	A105	LF2	F11	F304(L)	F316(L)	F51
15	Nameplate	AL	AL	AL	AL	AL	AL	AL
32	Revit	AL	AL	AL	AL	AL	AL	AL
33	Steel ball	430	430	304	STL	316(L)	316(L)	STL
34	Disc nut	2H	2H	8	8	8(M)	8(M)	8M
35	Hinge	410	410	304	410	316(L)	316(L)	F51
36	Pin	410	410	304	410	304(L)	316(L)	F51
37	Screwed stud	B7	B7	L7	B16	B8(M)	B8(M)	B8(M)
38	Nut	2H	2H	8	8	8(M)	8(M)	8(M)
39	Metal ring	304	304	304	304	304(L)	316(L)	F51

Female Threaded and Socket Welded Check Valves

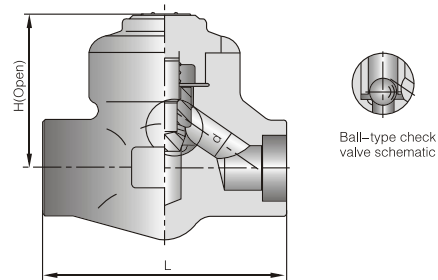
CL800 Bolted bonnet, full port and reducing port Threaded, butt-welded or socket welded ends; design to BS5352



To load the springs need to specify in the order form

Specification (NPS)	R.P			1/2	3/4	1	1¼	1½	2	2½
	F.P		1/4	3/8	1/2	3/4	1	1¼	1½	2
Face to face (mm)	L	Lift	79	79	92	111	120	152	172	220
		Swing	79	79	92	111	120	120	140	178
Heihgt (mm)	H	Lift	61	61	61	78	84	84	118	132
		Swing	61	61	61	78	84	84	120	133
Flow port dimension (mm)	d	Lift	7	9	13	17.5	23	30	35	46
		Swing	8	10.5	13.5	18	24	29	36.5	45
Weight (Kg)		Lift	1.2	1.5	1.7	3.3	4.2	4.2	10.5	12.5
		Swing	1.4	15	1.7	3.3	4.2	4.2	8.5	10.9

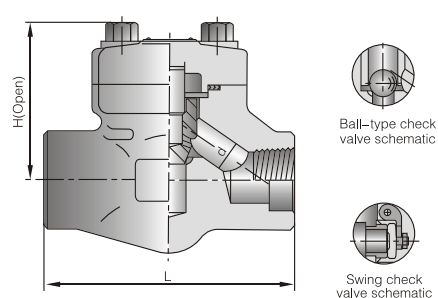
CL800 Welded bonnet, full port and reducing port Threaded, butt-welded or socket welded ends; design to BS5352



To load the springs need to specify in the order form

Specification (NPS)	R.P		1/2	3/4	1	1 ¹ / ₄	1 ¹ / ₂	2	2 ¹ / ₂
	F.P	1/4	3/8	1/2	3/4	1	1 ¹ / ₄	1 ¹ / ₂	2
Face to face (mm)	L	79	79	92	111	120	152	172	220
Heihgt (mm)	H	61	61	61	78	84	103	118	132
Flow port dimension (mm)	d	7	9	13	17.5	23	30	35	46
Weight (Kg)		1.2	1.3	1.5	3.0	3.9	6.0	10	12

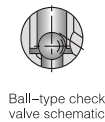
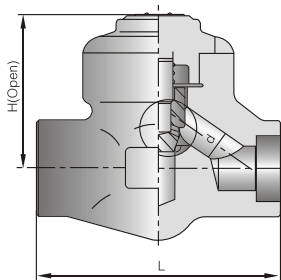
CL900-CL1500 Bolted bonnet, full port and reducing port Threaded, butt-welded or socket welded ends; design to BS5352



To load the springs need to specify in the order form

Specification (NPS)	R.P		1/2	3/4	1	1¼	1½	2	
	F.P		1/4	3/8	1/2	3/4	1	1½	2
Face to face (mm)	L	Lift	92	111	111	120	152	172	220
		Swing	92	111	111	120	120	140	178
Heihgt (mm)	H	Lift	61	78	78	84	103	118	132
		Swing	61	78	78	84	101	120	133
Flow port dimension (mm)	d	Lift	7	12	15	20	28	32	40
		Swing	8	10.5	13.5	187	24	29	45
Weight (Kg)		Lift	1.5	3.4	3.3	4.2	6.3	10.5	12.5
		Swing	1.5	3.4	3.3	4.2	5.0	8.5	10.9

Female Threaded and Socket Welded Check Valves



Ball-type check valve schematic

To load the springs need to specify in the order form

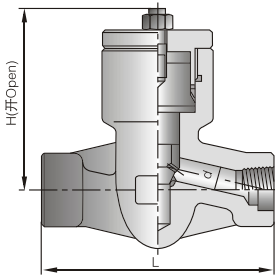
CL900–CL1500

Welded bonnet, full port and reducing port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	R.P		1/2	3/4	1	1¼	1½	2	
	F.P	1/4	3/8	1/2	3/4	1	1¼	1½	2
Face to face (mm)	L	92	111	111	120	152	172	220	
Heihgt (mm)	H	61	78	78	84	103	118	132	
Flow port dimension (mm)	d	7	12	15	20	28	32	40	
Weight (Kg)		1.3	3.1	3.1	3.9	5.8	10.0	11.5	

CL900–CL1500

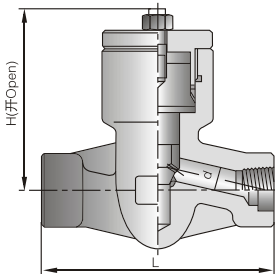
Pressure seal bonnet, full port and reducing port
Threaded, butt-welded or socket welded ends; design to BS5352



Specification (NPS)	R.P	–	1/2	3/4	1	1¼	1½	2	2½	3
	F.P	1/4	3/8	1/2	3/4	1	1¼	1½	2	2½
Face to face (mm)	L		140	140	140	178	216	216		
Heihgt (mm)	H		117	117	117	152	195	195		
Flow port dimension (mm)	d		12	15	20	28	28	40		
Weigh (Kg)			7.5	7.0	6.8	18.5	18.5	22		

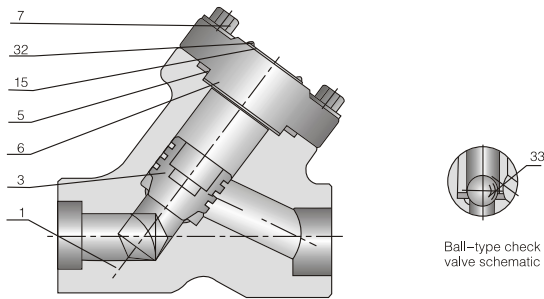
CL2500

Pressure seal bonnet, full port
Threaded, butt-welded or socket welded ends; design to ASME B16.34

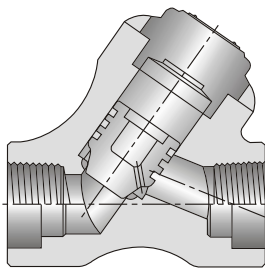


Specification (NPS)	F.P	1/4	3/8	1/2	3/4	1	1¼	1½	2
Face to face (mm)	L			186	186	186	232	232	279
Heihgt (mm)	H			117	117	117	152	150	195
Flow port dimension (mm)	d			11	14	14	25	28	35
Weigh (Kg)				11.8	11	11	23	26.4	39

Female Threaded Welded Y Type Check Valves



To load the springs need to specify in the order form



Application standards

- Design and manufacture conform to: BS5352; MSS SP-118
- Connection ends conform to:
 - Screw ends dimension conform to ANSI B16.11; JB/T1751
 - Screw ends dimension conform to ANSI B1.20.1; JB/T7306
 - Test and inspection conform to ANSI B16.25; JB/T12224
 - Structure features: ANSI B16.5; JB79
- Test and inspection conform to: API598; GB/T13927; JB/T9092
- Structure features: Bolted bonnet, welded bonnet
- Materials conform to ANSI/ASTM
- Main materials: A105; LF2; F5; F11; F22; 304(L); 316(L); F347; F321; F51; Monel; 20 Alloy

Carbon steel temperature–pressure rate

CL1500–3705 P.S.I @100° F

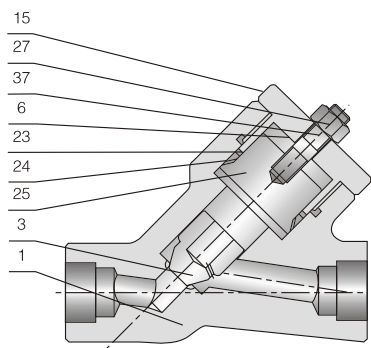
CL2500–6170 P.S.I @100° F

CL4500–1111 P.S.I @100° F

Main part materials

NO.	Part name	A105/F6a	A105/F6aHFS	LF2/304	F11/F6aHF	F304(L)/304(L)	F316(L)/316(L)	F51/F51
1	Body	A105	A105	LF2	F11+HF	F304(L)	F316(L)	F51
3	Disc	410	410HF	304	410HF	304(L)	316(L)	F51
5	Gasket	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	316+ Flexible graphite	316+ Flexible graphite
6	Bonnet	A105	A105	LF2	F11	F304(L)	F316(L)	F51
7	Bolt	B7	B7	L7	B16	B8(M)	B8(M)	B8M
15	Nameplate	AL	AL	AL	AL	AL	AL	AL
32	Revit	AL	AL	AL	AL	AL	AL	AL
33	Steel ball	403	403	304	STL	316(L)	316(L)	STL

Y Type Pressure Sealed Check Valves



Application standards

- 1、 Design and manufacture conform to: ASME B16.34; MSS SP-118
- 2、 Connection ends conform to:
 - 1) Screw ends dimension conform to ANSI B16.11; JB/T1751
 - 2) Screw ends dimension conform to ANSI B1.20.1; JB/T7306
 - 3) Test and inspection conform to ANSI B16.25; JB/T12224
 - 4) Structure features: ANSI B16.5; JB79
- 3、 Test and inspection conform to: API598; GB/T13927; JB/T9092
- 4、 Structure features:

A threaded and pressure seal bonnet; Y type and T type
- 5、 Materials conform to ANSI/ASTM
- 6、 Main materials: A105; LF2; F5; F11; F22; 304(L); 316(L); F347; F321; F51; Monel; 20 Alloy.

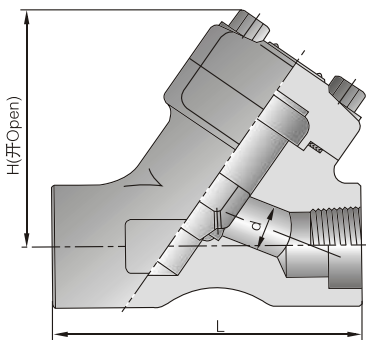
Carbon steel temperature–pressure rate

CL1500–3705 P.S.I @100° F
CL2500–6170 P.S.I @100° F
CL4500–1111 P.S.I @100° F

Main part materials

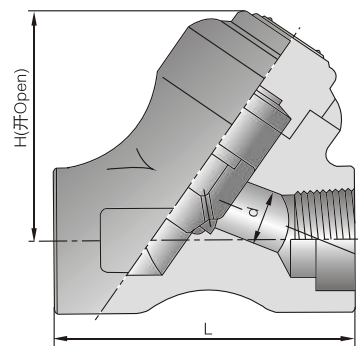
NO.	Part name	A105/F6a	A105/F6aHFS	LF2/304	F11/F6aHF	F304(L)/304(L)	F316(L)/316(L)	F51/F51
1	Body	A105	A105+HF	LF2	F11+HF	F304(L)	F316(L)	F51
3	Disc	F6a	F6a	304	F6aHF	F304(L)	F316(L)	F51
6	Bonnet	A105	A105	LF2	F11	F304(L)	F316(L)	F51
15	Nameplate	AL	AL	AL	AL	AL	AL	AL
23	Seal ring gasket	420	420	304	304	304(L)	316(L)	41
24	P.S.ring	304	304	304	304	316L	316L	316L
25	P.S.seat	F410	F410	F304	F410	F304	F316	F51
27	Lift nut	2H	2H	8	8	8(M)	8(M)	8M
37	Lift stud	B7	B7	L7	B16	B8(M)	B8(M)	B8M

Y Type Check Valves



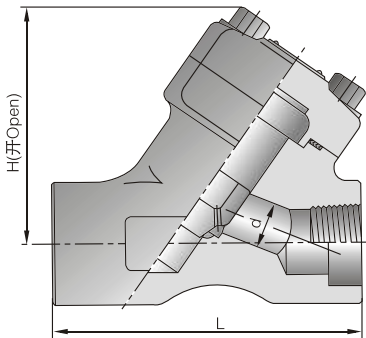
CL800 Bolted bonnet, full port and reducing port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	R.P	–	1/2	3/4	1	1¼	1½	2	2½	3
	F.P	1/4	3/8	1/2	3/4	1	1¼	1½	–	–
Face to face (mm)	L	98	98	98	120	140	140	170	–	–
Height (mm)	H	70	100	70	100	110	120	120	–	–
Flow port dimension (mm)	d	7	180	13	17.5	23	30	35	–	–
Weight (Kg)		2.2	9	2.1	4.2	9	8.9	10	–	–



CL800 Welded bonnet, full port and reducing port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	R.P	–	1/2	3/4	1	1¼	1½	2	2½	3
	F.P	1/4	3/8	1/2	3/4	1	1¼	1½	–	–
Face to face (mm)	L	98	98	98	120	140	140	170	–	–
Height (mm)	H	65	65	65	95	105	110	110	–	–
Flow port dimension (mm)	d	7	10	13	17.5	23	30	35	–	–
Weight (Kg)		1.8	1.8	2.0	3.5	8.0	8.0	12	–	–



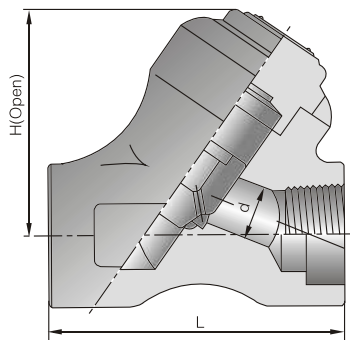
CL900–CL1500 Bolted bonnet, full port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	F.P	3/8	1/2	3/4	1	1¼	1½	2	2½
Face to face (mm)	L	120	120	120	140	170	170	220	–
Height (mm)	H	70	70	100	110	110	120	150	–
Flow port dimension (mm)	d	9	12	15	20	28	32	40	–
Weight (Kg)		2.1	4.2	9	8.9	10	18.6	20	–

Y Type Check Valves

CL900–CL1500

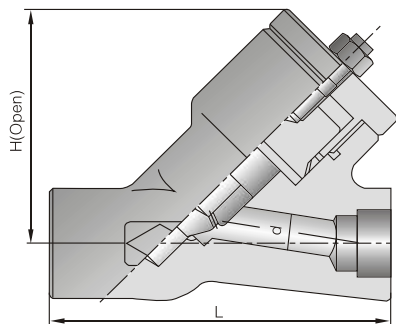
Welded bonnet, full port
Threaded, butt-welded or socket welded ends; design to BS5352



Specification (NPS)	F.P	3/8	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2
Face to face (mm)	L	120	120	120	140	170	170	220	–
Height (mm)	H	65	65	65	105	110	110	140	–
Flow port dimension (mm)	d	9	12	15	20	32	28	40	–
Weight (Kg)		2.0	3.5	3.5	8.0	12	12	18	–

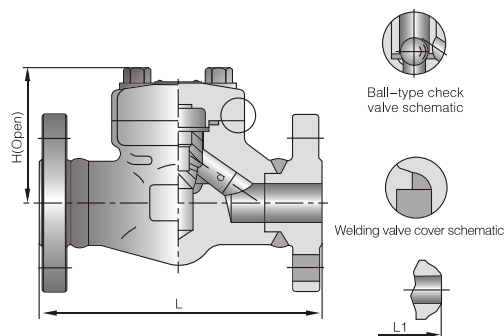
CL2500

Pressure seal, bolted bonnet, full port
Threaded, butt-welded or socket welded ends; design to ASME B16.34



Specification (NPS)	F.P	3/8	1/2	3/4	1	1 1/4	1 1/2	2
Face to face (mm)	L	186	186	186	186	232	232	310
Height (mm)	H	233	233	233	233	256	256	330
Flow port dimension (mm)	d	9	11	14	19	25	28	35
Weight (Kg)		11.2	11.5	10.6	10.8	25	22	39

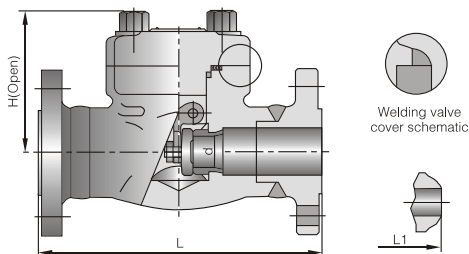
Flange and Butt-welded Check Valves



CL150–300–600

Bolted bonnet, full port
Flange–welded or butt–welded ends; design to BS5352

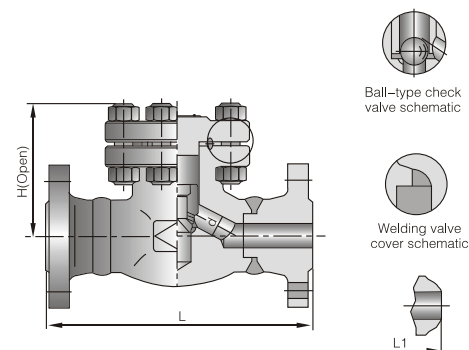
Specification (NPS)		F.P	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2	2
Face to face (mm)	CL150	L(RF) L1(BW)	–	–	108	118	127	140	165	203
	CL300		–	–	153	178	203	216	229	267
	CL600		–	–	165	191	216	229	241	292
Heihgt (mm)	CL150	H	–	–	77	81	93	95	103	118
	CL300/600		–	–	61	78	84	101	120	133
Flow port dimension (mm)		d	–	–	10	13	17.5	23	30	35
Weight (Kg)	CL150		–	–	3.6	4.6	8.5	9.2	12.5	14.8
			–	–	3.0	3.6	7.6	8.5	11.3	13.6
	CL300		–	–	3.7	4.8	8.8	9.6	13.7	17.8
			–	–	3.2	4.3	8.0	8.6	12.7	16.2
	CL600		–	–	4.0	5.8	9.5	10.4	15.6	24.5
			–	–	3.4	5.1	8.8	9.2	14.8	22.5



CL150–300–600

Bolted bonnet, full port
Flange–welded or butt–welded ends; design to BS5352

Specification (NPS)		F.P	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2	2
Face to face (mm)	CL150	L(RF) L1(BW)	–	–	108	118	127	140	165	203
	CL300		–	–	153	178	203	216	229	267
	CL600		–	–	165	191	216	229	241	292
Heihgt (mm)	CL150	H	–	–	77	81	93	95	103	118
	CL300/600		–	–	61	78	84	101	120	133
Flow port dimension (mm)		d	–	–	10	13.5	18	24	29	36.5
Weight (Kg)	CL150		–	–	3.6	4.6	8.5	9.2	12.5	14.8
			–	–	3.0	3.6	7.6	8.5	11.3	13.6
	CL300		–	–	3.7	4.8	8.8	9.6	13.7	17.8
			–	–	3.2	4.3	8.0	8.6	12.7	16.2
	CL600		–	–	4.0	5.8	9.5	10.4	15.6	24.5
			–	–	3.4	5.1	8.8	9.2	14.8	22.5

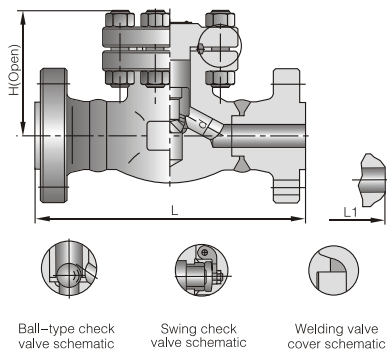


CL150–300–600

Bolted bonnet, full port
Flange–welded or butt–welded ends; design to BS5352

Specification (NPS)		F.P	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2	2
Face to face (mm)	CL150	L(RF) L1(BW)	–	–	108	118	127	140	165	203
	CL300		–	–	153	178	203	216	229	267
	CL600		–	–	165	191	216	229	241	292
Heihgt (mm)	CL150	H	–	–	77	81	93	95	103	118
	CL300/600		–	–	61	78	84	101	120	133
Flow port dimension (mm)		d	–	–	10	13	17.5	23	30	35
Weight (Kg)	CL150		–	–	3.2	3.5	4.6	5.2	7.0	16
			–	–	2.8	3.0	4.0	4.6	6.3	15
	CL300		–	–	4.6	6.1	9.1	12	16	21
			–	–	4.1	5.7	8.4	11.2	14.5	19.5
	CL600		–	–	4.8	6.3	9.3	13	16.5	22
			–	–	4.4	5.9	8.7	12.1	15.8	20.8

Flange and Butt-welded Check Valves



CL900–CL1500

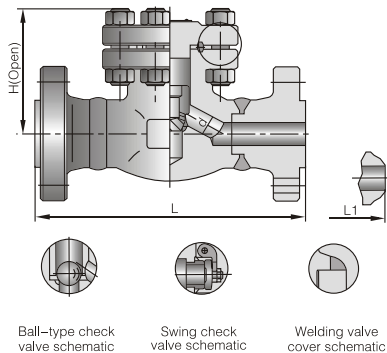
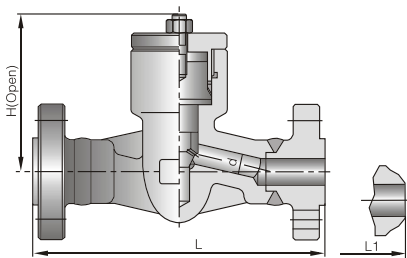
Bolted bonnet, full port
Flange–welded or butt–welded ends; design to BS5352

Specification (NPS)		1/2	3/4	1	1¼	1½	2
Face to face (mm)	L(RJ) L1(BW)	216	229	254	280	305	371
Heihgt (mm)	H	81	93	95	101	118	130
Flow port dimension (mm)	d						
	Lift	12	15	20	28	32	40
Weight (Kg)	Swing	13.5	18	24	29	36.5	45
	Lift	5.2	6.8	10.5	28	18	24
Weight (Kg)	Swing	5.0	6.1	10.8	29	17.6	27

CL900–CL1500

Pressure seal, bolted full port and reducing port
Flange–welded or butt–welded ends; design to BS5352

Specification (NPS)		3/8	1/2	3/4	1	1¼	1½	2
Face to face (mm)	L(RJ) L1(BW)	–	216	229	254	280	305	268
	L(RTJ)	–	216	229	254	280	305	371
Heihgt (mm)	H	–	117	117	117	152	152	195
Flow port dimension (mm)	d	–	12	15	20	28	32	40
Weight (Kg)		–	10.5	11.9	13.9	19.9	26.9	32.5



CL2500

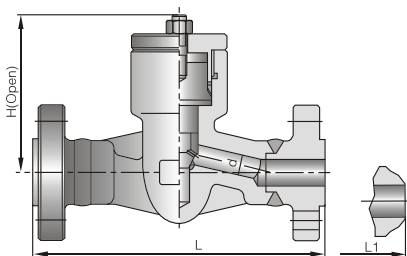
Bolted bonnet, full port
Flange–welded or butt–welded ends; design to BS5352

Specification (NPS)		1/2	3/4	1	1¼	1½	2
Face to face (mm)	L(RJ) L1(BW)	264	273	308	349	384	450
	L(RTJ)	264	273	308	352	387	454
Heihgt (mm)	H	81	93	95	101	118	130
Flow port dimension (mm)	d						
	Lift	12	15	20	28	32	40
Weight (Kg)	Swing	10.5	13.5	18	24	29	36.5
	Lift	17	21	28	14.5	58	85
Weight (Kg)	Swing	5.0	6.1	10.8	11.2	17.6	27

CL2500

Pressure seal, bolted full port and reducing port
Flange–welded or butt–welded ends; design to BS5352

Specification (NPS)		3/8	1/2	3/4	1	1¼	1½	2
Face to face (mm)	L(RJ) L1(BW)	–	264	273	308	349	384	450
	L(RTJ)	–	264	273	308	352	387	454
Heihgt (mm)	H	–	117	117	117	152	152	195
Flow port dimension (mm)	d	–	12	15	20	32	28	40
Weight (Kg)		–	12.6	14.9	16.5	24.8	30	35



Y-type Strainers

Construction is as follows

Full port or conventional port;

Bolted bonnet & spiral wound gasket seal bonnet;

Socket weld ends to ASME B16.11

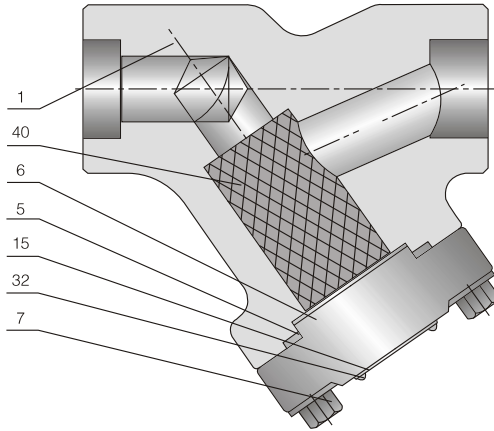
Screwed ends (NPT) to ANSI/ASME B1.20.1.

Equip blow down tap and renewable strainer.

Wanli Forged steel Y type strainer,
bolted bonnet, gasket adopt spiral
wound (304+flexible graphite) or
metalling seal.



Y-type Strainers



Application standards

- 1、Design and manufacture conform to: BS5352
- 2、Connection ends conform to:
 - 1) Screw ends dimension conform to ANSI B16.11; JB/T1751
 - 2) Screw ends dimension conform to ANSI B1.20.1; JB/T7306
 - 3) Test and inspection conform to ANSI B16.25; JB/T12224
 - 4) Structure features: ANSI B16.5; JB79
- 3、Test and inspection conform to: API598; GB/T13927; JB/T9092
- 4、Structure features:
Bolted bonnet or welded bonnet
- 5、Materials conform to ANSI/ASTM
- 6、Main materials: A105; LF2; F5; F11; F22; 304(L); 316(L); F347; F321; F51; Monel; 20 Alloy

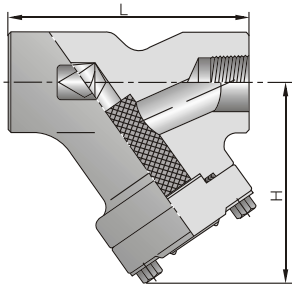
Carbon steel temperature–pressure rate

CL150–285 P.S.I @100° F
CL300–740 P.S.I @100° F
CL600–1480 P.S.I @100° F
CL800–1975 P.S.I @100° F
CL1500–3705 P.S.I @100° F
CL2500–6170 P.S.I @100° F

Main part materials

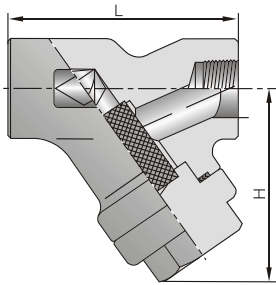
NO.	Part name	A105/F6a	A105/F6aHFS	LF2/304	F11/F6aHF	F304(L)/304(L)	F316(L)/316(L)	F51/F51
1	Body	A105	A105	LF2	F11	F304(L)	F316(L)	F51
5	Gasket	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite	304+ Flexible graphite
6	Bonnet	A105	A105	LF2	F11	F304(L)	F316(L)	F51
7	Bolt	B7	B7	L7	B16	B8(M)	B8(M)	B8M
15	Nameplate	AL	AL	AL	AL	AL	AL	AL
32	Rivet	H62	H62	H62	H62	H62	H62	H62
40	Filter screen	304	304	304	304	304(L)	316(L)	316(L)

Y-type Strainers



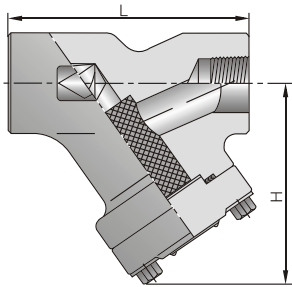
CL800 Bolted bonnet, full port and reducing port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	R.P	–	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3
	F.P	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2	–	–
Face to face (mm)	L	98	98	98	120	140	140	170	–	–
Height (mm)	H	70	70	70	100	110	120	120	–	–
Flow port dimension (mm)	d	7	9	13	17.5	30	30	35	–	–
Weight (Kg)		2.2	2.2	2.1	4.2	8.9	8.9	10	–	–



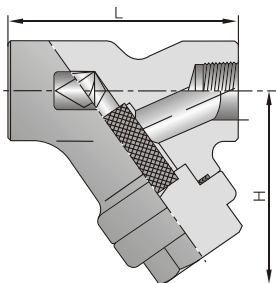
CL800 Welded bonnet, full port and reducing port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	R.P	–	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3
	F.P	1/4	3/8	1/2	3/4	1	1 1/4	1 1/2	–	–
Face to face (mm)	L	98	98	98	120	140	140	170	–	–
Height (mm)	H	65	65	65	95	105	110	110	–	–
Flow port dimension (mm)	d	7	9	13	17.5	23	30	35	–	–
Weight (Kg)		1.8	1.8	2.0	3.5	9	8.0	12	–	–



CL900–CL1500 Bolted bonnet, full port
Threaded, butt-welded or socket welded ends; design to BS5352

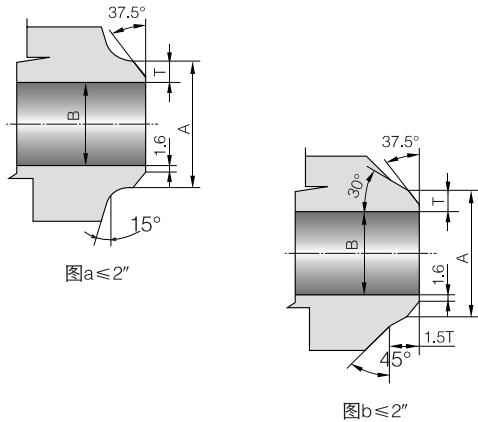
Specification (NPS)	F.P	3/8	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2
Face to face (mm)	L	120	120	120	140	170	170	220	–
Height (mm)	H	70	70	100	110	110	120	150	–
Flow port dimension (mm)	d	9	12	15	20	28	32	40	–
Weight (Kg)		2.1	4.2	9	8.9	10	18.6	20	–



CL900–CL1500 Welded bonnet, full port
Threaded, butt-welded or socket welded ends; design to BS5352

Specification (NPS)	F.P	3/8	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2
Face to face (mm)	L	120	120	120	140	170	170	220	–
Height (mm)	H	65	65	95	105	110	110	140	–
Flow port dimension (mm)	d	9	12	15	20	28	32	40	–
Weight (Kg)		2.0	3.5	8.0	8.0	12	16	18	–

End Connection Standard



Butt welding standard–ASME B16.25

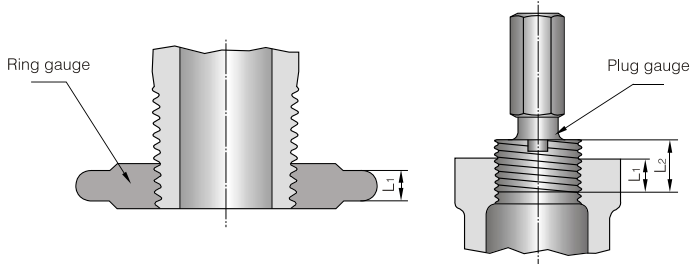
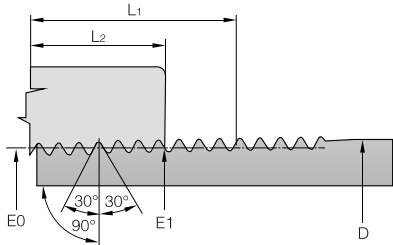
Unit: mm

in	SCH.40			SCH.80			SCH.160			SCH.XXS		
	A	B	T	A	B	T	A	B	T	A	B	T
1/2	21.3	15.8	2.77	21.3	13.8	3.73	21.3	11.7	4.78	21.3	6.4	7.47
3/4	26.7	25.0	2.87	26.7	18.9	3.91	26.7	15.6	5.56	26.7	11.1	7.82
1	33.4	26.6	3.38	33.4	24.3	4.55	33.4	20.7	6.35	33.4	15.2	9.09
1 1/4	42.2	35.1	3.55	42.2	32.5	4.85	42.2	29.5	6.35	42.2	22.8	9.70
1 1/2	48.3	41.0	3.68	48.3	38.1	5.08	48.3	33.5	7.41	48.3	28.0	10.15
2	60.3	52.5	3.91	60.3	49.2	5.54	60.3	42.9	8.74	60.3	38.2	11.07
2 1/2	73.3	62.7	5.15	73.0	59.0	7.01	73.0	54.0	9.53	73.0	45.0	14.02
3	88.9	78.0	5.48	88.9	73.7	7.62	88.9	66.6	11.13	88.9	58.4	15.24
4	114.3	102.3	6.02	114.3	97.2	8.56	114.3	85.3	13.49	114.3	80.1	17.12

All socket welding standard JB–ANSI–JIS

Unit: mm

in	JB		ANSI		JIS	
	D	L	D	L	D	L
1/4	14.5	10.0	14.1	9.6	14.1	9.6
3/8	18.0	10.0	17.6	9.6	17.9	9.6
1/2	22.5	10.0	21.8	9.6	22.2	9.6
3/4	28.5	11.0	27.1	12.7	27.7	12.7
1	34.5	12.0	33.8	12.7	34.5	12.7
1 1/4	43.0	14	42.6	12.7	43.2	12.7
1 1/2	49.0	15	48.7	12.7	49.1	12.7
2	61.2	16	61.2	15.9	61.2	15.9



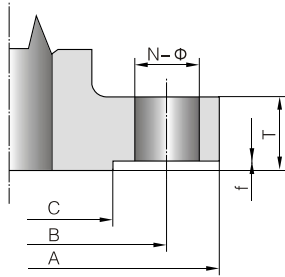
Taper thread plug gauge and ring gauge measurement type, the right way is to manually spin the timely and sufficient number of turns to avoid the complexity of use and product cases caused gauge down.

Thread standard–ASME B1.20.1

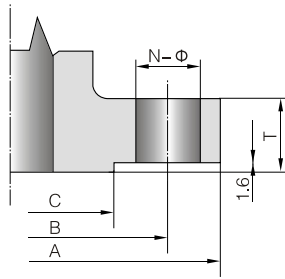
in	D	Teeth/inch	P	E0	E1	L1	L2	Tooth height
	Tube diameter		Pitch	Pipe thread diameter	Large diameter internal thread	Inside and outside the length of thread engagement	Effective length of thread	
1/16	7.93	27	0.94	6.89	7.41	4.06	6.63	0.73
1/8	10.29	27	0.94	9.23	9.49	4.10	6.70	0.73
1/4	13.72	18	1.41	12.13	12.49	5.79	10.20	1.13
3/8	17.15	18	1.41	15.55	15.93	6.10	10.36	1.13
1/2	21.34	14	1.81	19.26	19.77	8.13	13.56	1.45
3/4	26.67	14	1.81	24.58	25.12	8.61	13.86	1.45
1	33.40	11.5	2.21	31.46	31.45	10.16	17.34	1.77
1 1/4	42.16	11.5	2.21	39.55	40.22	10.67	17.95	1.77
1 1/2	48.26	11.5	2.21	45.62	46.29	10.67	18.38	1.77
2	60.33	11.5	2.21	57.63	58.33	11.07	19.22	1.77

Flange Size

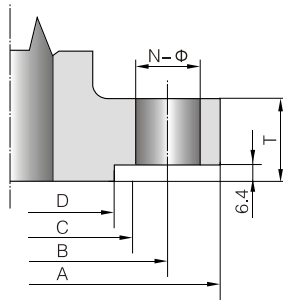
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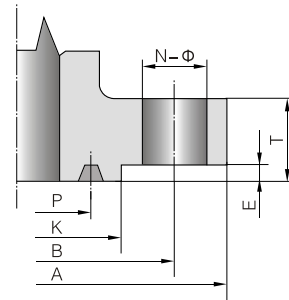
PN	DN	NPS	A	B	C	T	f	Bolt hole	
								N	Φ
25 40	15	1/2	95	65	45	16	2	4	14
	20	3/4	105	75	58	18	2	4	14
	25	1	115	85	68	18	2	4	14
	32	1 1/4	140	100	78	18	2	4	18
	40	1 1/2	150	110	88	18	2	4	18
	50	2	165	125	102	20	2	4	18
64	15	1/2	105	75	45	20	2	4	14
	20	3/4	130	90	58	22	2	4	18
	25	1	140	100	65	24	2	4	18
	32	1 1/4	155	110	75	24	2	4	22
	40	1 1/2	170	125	88	26	3	4	22
	50	2	180	135	95	26	3	4	22



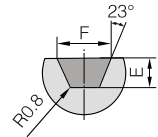
RF 150~300Lb



RF 600~2500Lb



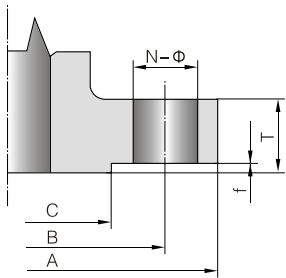
RJ 600~2500Lb



ASME-B16.5

CLASS	NPS	A	B	C	T	Bolt hole		Ring joint face				Ring number
						N	Φ	K	P	F	E	
150	1/2	89	60.3	34.9	11.5	4	16	—	—	—	6.4	—
	3/4	98	69.8	42.9	13.0	4	16	—	—	—	6.4	—
	1	108	79.4	50.8	14.5	4	16	63.5	47.62	8.74	6.4	R15
	1 1/4	117	88.9	63.5	16.0	4	16	73.0	57.15	8.74	6.4	R17
	1 1/2	127	98.4	73.0	17.5	4	16	82.5	65.07	8.74	6.4	R19
	2	152	120.6	92.1	19.5	4	20	102.0	82.55	8.74	6.4	R22
300	1/2	95	66.7	34.9	14.5	4	16	51.0	34.14	7.14	6.4	R11
	3/4	117	82.5	42.9	16.0	4	20	63.5	42.88	8.74	6.4	R13
	1	124	88.9	50.8	17.5	4	20	70.0	50.80	8.74	6.4	R16
	1 1/4	133	98.4	63.5	19.5	4	20	79.5	60.32	8.74	6.4	R18
	1 1/2	156	114.3	73.0	21.0	4	23	90.5	68.28	8.74	6.4	R20
	2	165	127.0	92.1	22.5	4	20	108.0	82.55	11.91	8.0	R23
600	1/2	95	66.7	34.9	14.5	4	16	51.0	34.14	7.14	5.6	R11
	3/4	117	82.5	42.9	16.0	4	20	63.5	42.88	8.74	6.4	R13
	1	124	88.9	50.8	17.5	4	20	70.0	50.80	8.74	6.4	R16
	1 1/4	133	98.4	63.5	21.0	4	20	79.5	60.32	8.74	6.4	R18
	1 1/2	156	114.3	73.0	22.5	4	23	90.5	68.28	8.74	6.4	R20
	2	165	127.0	92.1	25.5	8	20	108.0	82.55	11.91	8.0	R23
900 1500	1/2	121	82.5	34.9	22.5	4	23	60.5	39.67	8.74	6.4	R12
	3/4	130	88.9	42.9	25.5	4	23	66.5	44.45	8.74	6.4	R14
	1	149	101.6	50.8	29.0	4	26	71.5	50.80	8.74	6.4	R16
	1 1/4	159	111.1	63.5	29.0	4	26	81.0	60.32	8.74	6.4	R18
	1 1/2	178	123.8	73.0	32.0	4	29	92.0	68.28	8.74	6.4	R20
	2	216	165.1	92.1	38.5	8	26	124.0	95.25	11.91	8.0	R24
2500	1/2	133	88.9	34.9	30.5	4	23	65.0	42.88	8.74	6.4	R13
	3/4	140	95.2	42.9	32.0	4	23	73.0	50.80	8.74	6.4	R16
	1	159	107.9	50.8	35.0	4	26	82.5	60.32	8.74	6.4	R18
	1 1/4	184	130.2	63.5	38.5	4	29	101.5	72.24	11.91	6.4	R21
	1 1/2	203	146	73.0	44.5	4	32	114.5	82.55	119.1	6.4	R23
	2	235	171	92.1	51.0	8	29	133.5	101.60	119.1	8.0	R26

Flange Size



RF
10K~63K

JIS Flange–JB/T79.1~4–94; JB/T82.1~4–94

PN (MPa)	DN	A	B	C	T	f	Bolt hole	
							N	Φ
10K	10	90	65	48	12	1	4	15
	15	95	70	52	12	1	4	15
	20	100	75	58	14	1	4	15
	25	125	90	70	14	1	4	19
	32	135	100	80	16	2	4	19
	40	140	105	85	16	2	4	19
	50	155	120	100	16	2	4	19
20K	10	90	65	46	14	1	4	15
	15	95	70	52	14	1	4	15
	20	100	75	58	16	1	4	15
	25	125	90	70	16	1	4	15
	32	135	100	80	18	2	4	19
	40	140	105	85	18	2	4	19
	50	155	120	100	18	2	8	19
30K	10	110	75	52	16	1	4	19
	15	115	80	55	18	1	4	19
	20	120	85	60	18	1	4	19
	25	130	95	70	20	1	4	19
	32	140	105	80	22	2	4	19
	40	160	120	90	22	2	4	23
	50	165	130	105	22	2	8	19
40K	10	110	57	52	18	1	4	19
	15	115	80	55	20	1	4	19
	20	120	85	60	20	1	4	19
	25	130	95	70	22	1	4	19
	32	140	105	80	24	2	4	19
	40	160	120	90	24	2	4	23
	50	165	130	105	26	2	8	19
63K	10	115	80	52	23	1	4	19
	15	120	85	55	23	1	4	19
	20	135	95	60	25	1	4	23
	25	140	100	70	27	1	4	23
	32	150	110	80	30	2	4	23
	40	175	130	90	32	2	4	25
	50	185	145	105	34	2	8	23

Commonly Used Materials for Chemical and Physical Properties of ASTM

Body and cover material

Chemical composition	Carbon steel	Low temperature		Alloy steel				Austenitic stainless steel ASTM-A182						Duplex stainless steel
	A105	LF2	LF3	F5	F91	F11Class2	F11Class3	F304	F304H	F304L	F316	F316L	F347H	F51
C	0.35	0.35	0.20	0.15	0.08–0.12	0.10–0.20	0.05–0.15	0.08	0.04–0.10	0.035	0.08	0.035	0.04–0.1	0.030
Mn	0.60–1.05	0.60–1.35	0.90	0.30–0.60	0.30–0.60	0.30–0.80	0.30–0.60	2.00	2.00	2.00	2.00	2.00	2.00	2.00
P	0.035	0.035	0.035	0.030	0.020	0.040	0.040	0.045	0.045	0.045	0.045	0.045	0.045	0.030
S	0.040	0.040	0.040	0.030	0.010	0.040	0.040	0.030	0.030	0.030	0.030	0.030	0.030	0.20
Si	0.10–0.34	0.15–0.30	0.20–0.35	0.50	0.20–0.50	0.50–1.00	0.50	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Ni	0.40	0.40	3.3–3.7	0.50	0.40	–	–	8.0–11.0	8.0–11.0	8.0–13.0	10.0–14.0	10.0–15.0	9.0–13.0	4.5–6.5
Cr	0.30	0.30	0.30	4.0–6.0	8–9.5	1.00–1.50	2.00–2.50	18.0–20.0	18.0–20.0	18.0–20.0	16.0–18.0	16.0–18.0	17.0–20.0	21–23
Mo	0.12	0.12	0.12	0.44–0.65	0.44–0.65	0.44–0.65	0.87–1.13	–	–	–	2.00–3.00	2.00–3.00	–	2.5–3.5

Mechanical properties	ASTM A105	LF2	LF3	F5	F91	F11Class2	F11Class3	F304	F304H	F304L	F316	F316L	F347H	F51
≥ Tensile strength	70	70–95	70–95	70	85	70	70	75	75	70	75	70	70	90
≥ Yield strength	36	36	37.5	40	60	40	45	30	30	25	30	25	30	65
Extension rate	22	22	22	20	20	20	20	30	30	30	30	30	30	25
Section shrinkage	30	30	35	35	40	30	30	50	50	50	50	50	50	45
Brinell hardness	187(2)	197	197	143–217	≤248	143–207	156–207	–	–	–	–	–	–	–

Trim and bolting material

Chemical composition	Internal materials					Bolt material				
	AISI 410	AISI 416	AISI 420	ASTM B164 Monel	Stellite Gr.6	ASTM A193		AISI 430	ASTM A194	
						B7	B8		2H	G8
C	0.15max.	0.15max.	0.15max.	0.3max.	1.00	0.37–0.49	0.08max.	0.12max.	0.10min.	0.08max.
Mn	1.00max.	1.25max.	1.00max.	2.0max.	1.00max.	0.65–1.10	2.0max.	1.00max.	1.00max.	2.00max.
P	0.040	0.060max.	0.040	–	–	0.035	0.045	0.040	0.040max.	0.045
S	0.030	0.15max.	0.030	0.024	–	0.04	0.30	0.030	0.050max.	0.030
Si	1.00max.	1.00max.	1.00max.	0.5max.	1.00	0.15–0.35	1.00max.	1.00max.	0.40max.	1.00max.
Cr	11.50–13.50	12.0–14.0	12.0–14.0	–	28.00	0.75–1.20	18.0–20.0	14.0–18.0	–	18.0–20.0
Ni	–	–	–	63.0max.	3.0max.	–	8.00–11.0	–	–	8.00–11.0
Mo	–	0.600.max.	–	–	–	0.15–0.25	–	–	–	–
Cu	–	–	–	28.0–34.0	–	–	–	–	–	–
Other elements	–	–	–	Fe:2.5max.	Fe:3.0max. W:4.0 Co:balance	–	–	–	–	–

Mechanical properties	410	416	420	ASTM A164	Gr.6	B7	B8	430	2H	G8
≥ Tensile strength	99/85 70/130	85/170 85/170	149/298 105/210	70(20 49.2	– –	125 87.8	75 52.7	75.4 53	– –	– –
≥ Yield strength	59/170 42/120	59/158 42/90	119/199 84/140	25(20 17.6	– –	105 73.8	30 21	40 28	– –	– –
Extension rate	(15)(1)	(10)(1)	(8)(1)	(35)(2)	–	16	30	28	–	–
Section shrinkage	50/75	8/60	5/40	–	–	50	50	65	–	–
Brinell hardness	180–375	180/375	300–600	–	HRC min.37	–	–	160	248–352	126–300